

SeatRelay: Operator-Authorized Bus Seat Resale

3-Minute Video Demo & Architecture Script

Built for First Commit (Bharat Builds Tour, WeMakeDevs × AWS) | Target Duration: 2m 45s – 3m 00s

Submission Requirements Checklist Met

- ✓ **About the project:** Real problem of 100% last-minute cancellation loss vs. sold-out routes in India.
- ✓ **Tech stack & architecture:** Full-stack React + TypeScript + AWS Serverless architecture.
- ✓ **How AWS is used:** Step Functions state machine, DynamoDB atomic conditional writes, EventBridge, S3, Cognito.
- ✓ **Novelty & Innovation:** RBI e-Rupee (CBDC) programmable token escrow with T+0 instant multi-party settlement.
- ✓ **Learning & growth:** Sovereign India-Stack integration (DigiLocker + e-Rupee) alongside AWS event-driven systems.

Script Timeline Overview

Time	Section	Core Topic
0:00 – 0:40	Part 1: Project Overview & Problem	Fare waste vs. sold-out buses; why open forums fail
0:40 – 1:30	Part 2: End-to-End Live Product Demo	Seller release → DigiLocker & e-Rupee claim → Operator 1-click reissue
1:30 – 2:20	Part 3: Tech Stack & AWS Cloud	Step Functions, DynamoDB conditional writes, EventBridge, Cognito
2:20 – 2:45	Part 4: Novelty & India Stack	RBI e-Rupee (CBDC) programmable smart contract escrow (T + 0 payout)
2:45 – 3:00	Part 5: Key Learnings & Conclusion	Regulatory compliance (DPDP), transit security, closing summary

Detailed Video Script

Part 1: The Problem & Project Overview (0:00 – 0:40)

Visual & On-Screen Action

Open on the live SeatRelay landing page. Show the headline **“SeatRelay: Your seat doesn’t have to go to waste”**. Hover over a search result showing **“Sold Out”** next to a 100% cancellation penalty window on a carrier ticket.

Spoken Script:

“Namaste! In India, millions of intercity bus seats go to waste every week. When a traveller’s plans change close to departure, cancellation charges hit 100%—a complete loss of their fare. At the exact same minute, another passenger searching that identical route sees ‘Sold Out’.

Why can't travellers just give their ticket to someone else? Because tickets carry government photo IDs! Open ticket reselling fails: buyers get turned away at boarding, tickets get fraudulently sold twice, or scalpers inflate prices.

*Introducing **SeatRelay**: India's first operator-authorized bus seat resale protocol. We enable confirmed tickets to change hands at **exact face value** up to departure, with zero scalping, cryptographic identity verification, and authentic manifest reissuance directly on the operator's official manifest."*

Part 2: Three-Persona End-to-End Walkthrough (0:40 – 1:30)

Visual & On-Screen Action

Walk through the 3 personas in real time:

1. **Seller (Rahul)**: In *My Journeys*, click **"Release this seat"** on Bangalore → Chennai. Show Rs. 850 refund preview.
2. **Buyer (Priya)**: Search route, find berth U12, initiate **DigiLocker Aadhaar e-KYC**, and select **e-Rupee (CBDC)** checkout.
3. **Operator (SwiftBus)**: In *Dispatch*, show side-by-side manifest change and click **"Approve Reissue"**.

Spoken Script:

"Here is how the protocol functions in real life across three personas:

*1. **The Seller**: Rahul holds confirmed seat U12 on a Bangalore to Chennai sleeper at Rs. 850. Two hours before departure, standard cancellation gives him Rs. 0. He clicks **'Release this seat'**. His ticket remains valid until someone buys it.*

*2. **The Buyer**: Priya searches the route, finds Rahul's released berth at the exact Rs. 850 face-value fare. Before booking, she verifies her legal identity in one tap via our integrated **DigiLocker Aadhaar e-KYC gateway**. At checkout, she selects **RBI e-Rupee (CBDC)**, locking the fare into a programmable smart contract escrow.*

*3. **The Operator**: In the SwiftBus dispatch console, the operator sees a real-time manifest comparison: Rahul off, Priya on with verified ID. With **one click**, the operator signs the reissue. Rahul's ticket is invalidated, Priya receives an authentic encrypted QR boarding pass, and Rahul gets an instant, automated refund."*

Part 3: Tech Stack & AWS Cloud Architecture (1:30 – 2:20)

Visual & On-Screen Action

Open the **AWS Reference Architecture Modal** or highlight the architecture block diagram showing Step Functions, DynamoDB, EventBridge, Lambda, Cognito, and S3.

Spoken Script:

"Under the hood, SeatRelay is built on an enterprise-grade AWS Serverless architecture designed for zero data loss and massive holiday demand spikes:"

- **AWS Amplify Hosting & Amazon Cognito**: Powers our responsive React, TypeScript, and Tailwind frontend with role-based access control for travellers and fleet operators.

- **Amazon API Gateway & AWS Lambda:** RESTful serverless microservices executing atomic listing, search, and settlement validations.
- **Amazon DynamoDB Atomic One-Buyer Guarantee:** Built using conditional expressions (`attribute_exists(listingId) AND status = 'LISTED'`). When 50 buyers tap Claim in the same millisecond on a festival weekend, DynamoDB guarantees strictly **one** write succeeds. The other 49 receive a clean 409 Conflict with zero accidental charges.
- **AWS Step Functions Centerpiece:** Coordinates the transfer state machine. It enforces the 10-minute checkout window, pauses on a task token awaiting operator approval, and orchestrates atomic rollbacks if declined.
- **Amazon EventBridge Scheduler:** Spawns precision one-shot timers per listing, auto-expiring unsold seats at $T - 60$ minutes so the seller retains standard operator cancellation rights.
- **Amazon S3 & SNS:** Manages encrypted QR boarding passes and distributes real-time SMS/email alerts at every state transition.

Part 4: Novelty — Sovereign India Stack & RBI e-Rupee Escrow (2:20 – 2:45)

Visual & On-Screen Action

Switch to the **RBI e-Rupee Programmable Escrow Explorer** tab in the Live Portal. Highlight the contract address, SHA-256 escrow lock hash, and the “**T+0 Settled**” instant multi-party split badge.

Spoken Script:

*“Our primary technological novelty addresses India’s refund friction. Traditional bank refunds take 2 to 5 business days. SeatRelay integrates **RBI Central Bank Digital Currency (CBDC / e-Rupee) Programmable Smart Contracts**.*

*When the buyer checks out, e-Rupee tokens are encumbered with a strict programmability rule: unlock only upon operator cryptographic authorization. The millisecond the operator approves, an **atomic multi-party split executes at T+0 real-time speed**—delivering an instant face-value refund to the seller’s wallet with zero intermediary delay.”*

Part 5: Learnings & Closing Summary (2:45 – 3:00)

Visual & On-Screen Action

Return to the main dashboard or full-screen presenter view with GitHub repository URL and deployment links displayed on screen.

Spoken Script:

“Building SeatRelay gave us deep hands-on experience orchestrating AWS distributed systems under stringent DPDP Act 2023 privacy rules and real-world transit constraints.

*With SeatRelay: **Nobody loses their fare, nobody misses the bus, and every passenger on board is on the record. Thank you!**”*